ADDITIONAL PROJECT : UG3

Daily 1km High Resolution PM2.5 estimation in India(2000-2025)

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Introduction/Overview

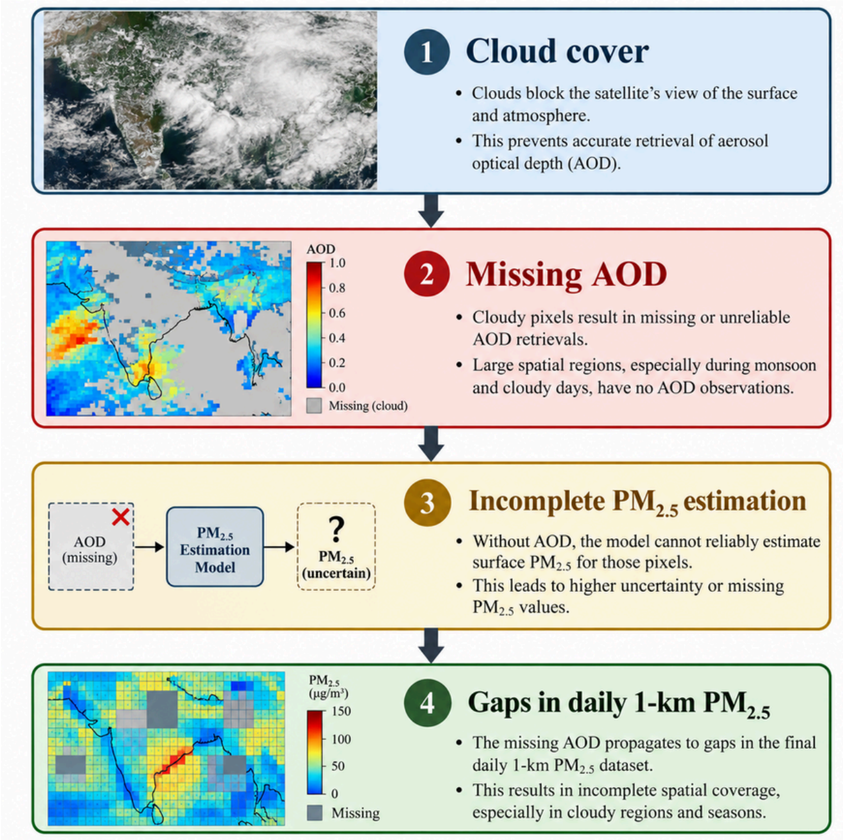
Problem Statement: why high resolution $PM_{2.5}$ data is needed in India?

- $PM_{2.5}$ is a major air pollutant associated with significant adverse health effects.
- Understanding its concentration and spatial distribution is important for air-quality management and exposure assessment.
- India has large spatial and temporal variations in $PM_{2.5}$ due to differences in:
  → Emission sources
  → Urbanisation
  → Geography
  → Meteorology
- Ground-based monitoring provides direct $PM_{2.5}$ measurements, but monitoring stations are:
  → Spatially limited
  → Unevenly distributed
  → Concentrated mainly in urban regions
- Ground observations therefore cannot provide continuous $PM_{2.5}$ information across the entire country.
- This creates a need for a: Daily + India-wide + High-resolution $PM_{2.5}$ dataset
- Previous work has demonstrated the feasibility of producing daily 1-km $PM_{2.5}$ over India using satellite and atmospheric reanalysis data.

FROM AOD TO $PM_{2.5}$: THE MISSING-DATA PROBLEM

Satellite AOD as a Source of High-Resolution $PM_{2.5}$:

- AOD represents the aerosol loading through the atmospheric column and can be used with atmospheric information to estimate surface $PM_{2.5}$.
- MODIS MAIAC provides daily Aerosol Optical Depth (AOD) at 1-km resolution.
- Satellite observations can provide spatial information beyond the limited coverage of ground monitoring stations.
- Therefore, satellite AOD provides an opportunity to develop high-resolution $PM_{2.5}$ maps over India.
- However, MAIAC is an optical satellite retrieval and is affected by cloud cover.
- Under cloudy conditions, AOD retrievals become unavailable, creating substantial spatial and temporal gaps.
- Our approach: Impute missing AOD → Complete daily 1-km AOD → Convert AOD to $PM_{2.5}$



LITERATURE REVIEW

A Satellite-Based High-Resolution (1-km) Ambient $PM_{2.5}$ Database for India over Two Decades (2000–2019)

Objective:

- Develop a continuous daily 1-km surface $PM_{2.5}$ dataset across India for 2000–2019.
- Analyse long-term $PM_{2.5}$ exposure and spatial pollution patterns.

Input Data :

- **MODIS MAIAC AOD:** 1-km resolution
- **CPCB $PM_{2.5}$:** 120 monitoring stations, 2009–2019
- **MERRA-2** aerosol diagnostics: 0.5° × 0.625°
- **GHSL:** urban/rural settlement information

**Limitation :

- MAIAC AOD experiences severe cloud-related data loss during the Southwest Monsoon.
- Missing AOD therefore limits the completeness of the daily $PM_{2.5}$ product.

Methodology:

- Derive a dynamic conversion factor:
- $\eta = PM_{2.5} / AOD$
- Estimate η using MERRA-2 aerosol vertical profiles and species composition.
- $PM_{2.5} = \eta \times AOD$ (MAIAC)
- Calibrate the MERRA-2-based relationship using CPCB observations.
- Apply Diurnal Scaling Factors (DSF) to convert satellite overpass concentrations to 24-hour daily $PM_{2.5}$.

Output :

- Daily 1-km $PM_{2.5}$ estimates across India for 2000–2019.

LITERATURE REVIEW

Spatiotemporal Imputation of MAIAC AOD Using Deep Learning with Downscaling (California - (2000–2016))

Objective:

- Reconstruct missing MAIAC AOD caused by incomplete satellite retrievals.
- Generate a more complete high-resolution AOD dataset.

Input Data :

- MAIAC AOD: 1-km spatial resolution
- Temporal resolution: weekly
- MERRA-2 GMI column AOD: 0.5° × 0.625°
- Meteorological variables, elevation and spatial coordinates
- AERONET AOD for validation

**Limitation :

- Imputation performed at weekly rather than daily resolution.
- Framework was developed and evaluated over California.
- Performance depends partly on pre-calculated monthly MAIAC information.

Methodology:

- Downscale coarse MERRA-2 AOD to 1-km resolution.
- Use an autoencoder-based deep residual network for AOD imputation.
- Apply residual connections, parameter sharing and bootstrap aggregation.
- Use AERONET for bias correction and validation.

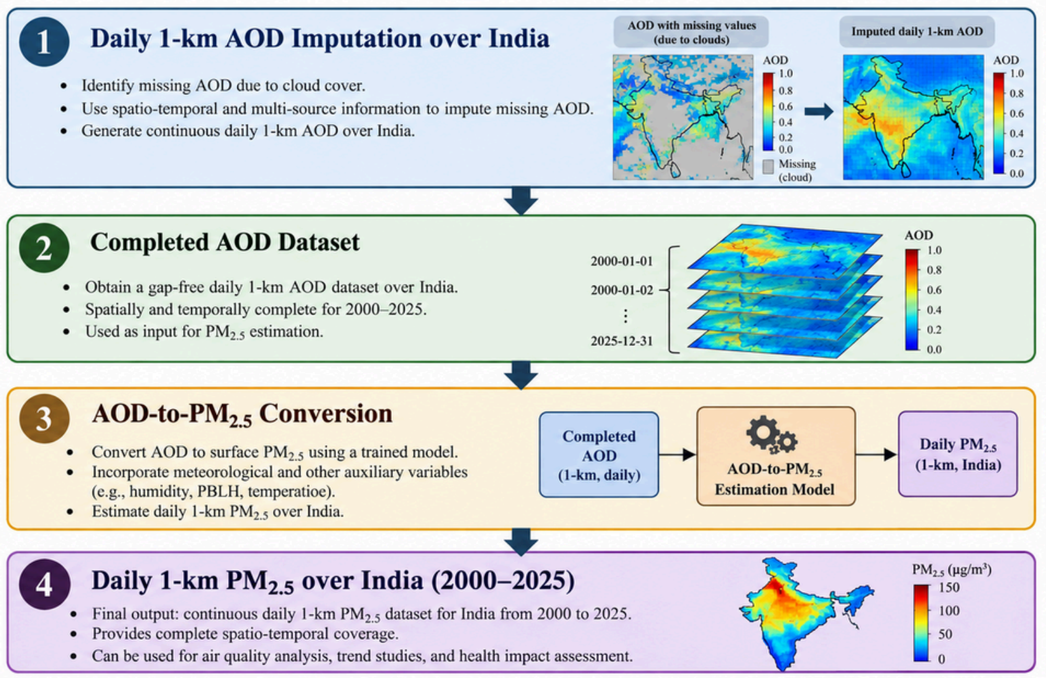
Output :

- Reconstructed weekly missing MAIAC AOD at 1-km spatial resolution.

IDENTIFIED RESEARCH GAP

Objective:

- Existing high-resolution $PM_{2.5}$ estimation over India relies on MAIAC AOD, but cloud-induced AOD gaps limit the spatial and temporal continuity of the resulting $PM_{2.5}$ product.
- Deep-learning approaches have demonstrated the feasibility of MAIAC AOD imputation at 1-km resolution, but the existing implementation is based on weekly data and a non-Indian study region.
- Therefore, there is a need for a daily, India-focused framework for reconstructing cloud-missing MAIAC AOD at 1-km resolution.
- Such reconstructed AOD can provide a more continuous input for daily 1-km $PM_{2.5}$ estimation over India.



OBJECTIVES

Main Objective:

- To develop a framework for generating a continuous daily 1-km $PM_{2.5}$ dataset over India for 2000–2025 by reconstructing cloud-induced gaps in satellite AOD.

Specific Objectives:

1)Daily 1-km AOD Imputation

- Develop a framework to reconstruct cloud-missing MAIAC AOD over India.
- Extend existing AOD imputation approaches from weekly to daily temporal resolution.
- Integrate relevant meteorological, aerosol and spatial information for AOD reconstruction.

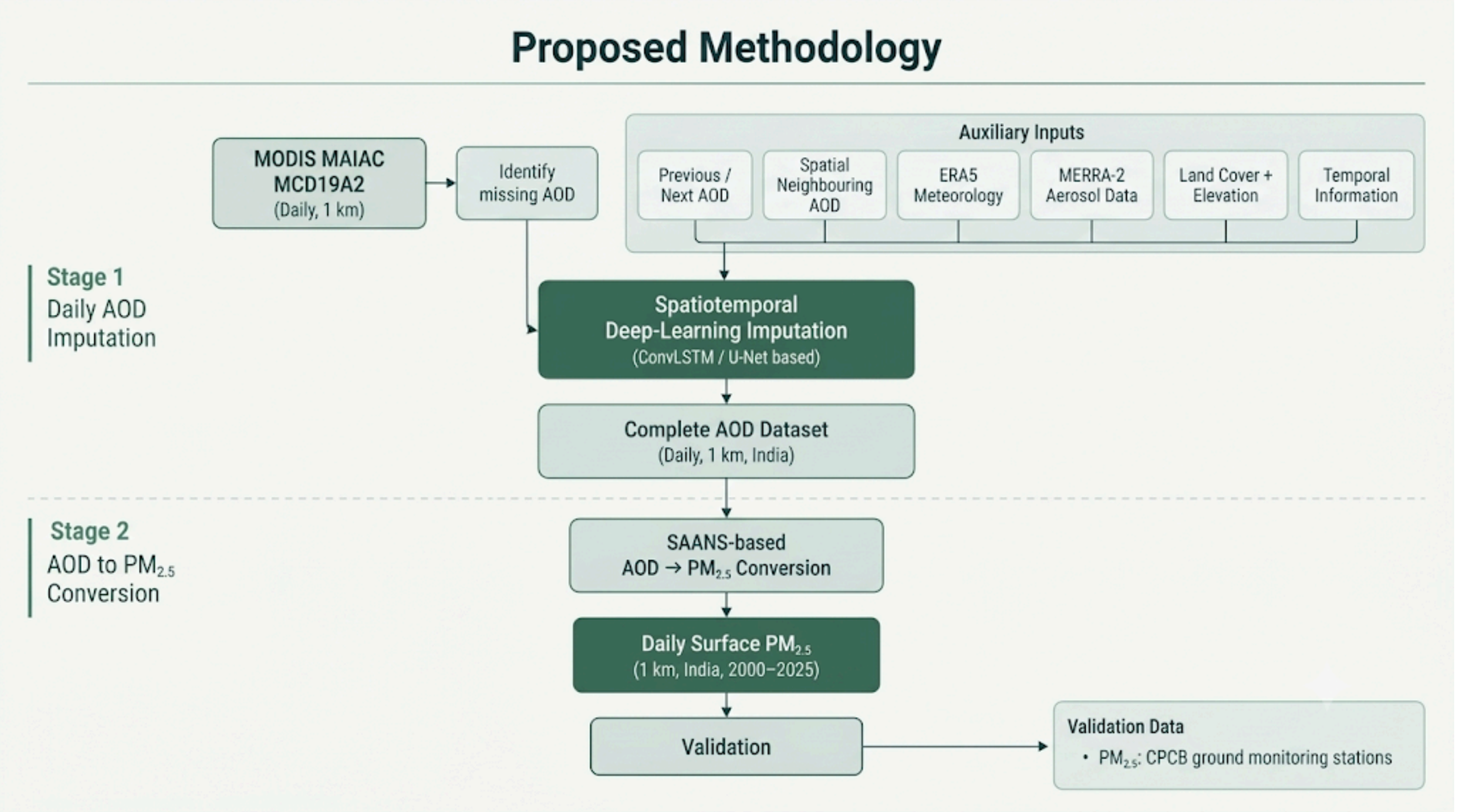
2)$PM_{2.5}$ Estimation

- Apply an established physical AOD-to-$PM_{2.5}$ conversion methodology to the reconstructed AOD.
- Generate daily 1-km surface $PM_{2.5}$ estimates over India for 2000–2025.

3). Validation and Evaluation

- Evaluate the accuracy of the reconstructed AOD using available independent observations.
- Assess the resulting daily 1-km $PM_{2.5}$ estimates using available ground-based $PM_{2.5}$ observations.

METHODOLOGY



DATA SOURCES AND AVAILABILITY

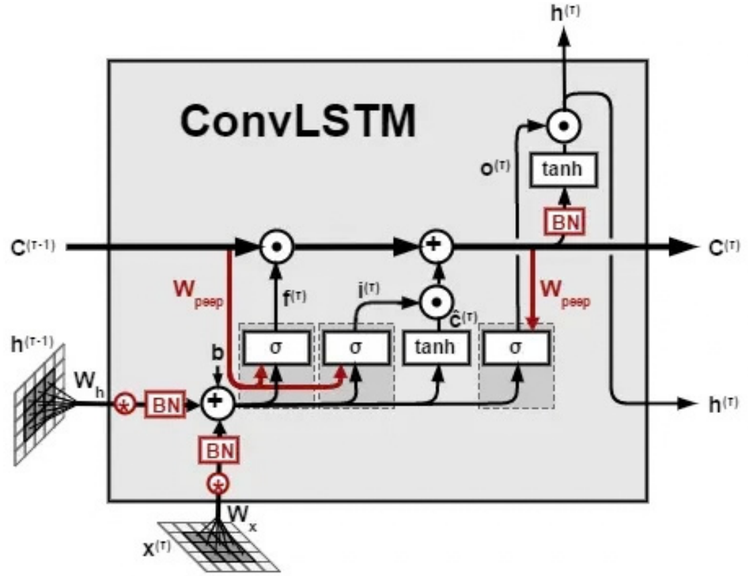
Dataset	Provider & Coverage	Resolution	Period	Key Information
MODIS MAIAC MCD19A2	NASA Earthdata / LP DAAC; Global land	1 km; Daily	2000–Present	Aerosol Optical Depth (AOD) at 550 nm; primary satellite AOD dataset
ERA5 Reanalysis	Copernicus C3S / ECMWF; Global	~0.25°; Hourly	1940–Present	Meteorological variables such as temperature, wind, humidity and precipitation
MERRA-2	NASA GMAO / GES DISC; Global	0.5° × 0.625°; Hourly	1980–Present	Aerosol diagnostics, composition and atmospheric information
AERONET	NASA AERONET; Global stations	Station-based	1993–Present	Ground-based AOD observations for independent validation
CPCB $PM_{2.5}$	CPCB, India; Monitoring stations	Station-based	Station-dependent	Ground-based $PM_{2.5}$ observations for validation

METHODOLOGY

ConvLSTM:

→ In this kind of architecture, the model passes the previous hidden state to the next step of the sequence. Therefore holding information on previous data the network has seen before and using it to make decisions.

→ When working with images, the best approach is a CNN (Convolutional Neural Network) architecture. The image passes through Convolutional Layers, in which several filters extract important features. After passing some convolutional layers in sequence, the output is connected to a fully-connected Dense network.



SAANS-based (AOD → $PM_{2.5}$) :

→ Derive a dynamic conversion factor:
$\eta = PM_{2.5} / AOD$
→ Instantaneous PM2.5 = Scaling Factor × MAIAC AOD i.e [$PM2.5=\eta \times AOD(MAIAC)$]
→ Diurnal Scaling Factor (DSF):
DSF = 24-h PM2.5 / Overpass-time PM2.5
→ Daily PM2.5=DSF × Instantaneous satellite PM2.5

Validation:

→ Validating imputed AOD using AERONET
→ Validating PM2.5 concentration using CPCB

FUTURE WORK PLAN

PHASE 1:

DATA PREPARATION

- Acquire daily MAIAC AOD for India
- Preprocess and quality-control input datasets
- Harmonize meteorological, aerosol and spatial data
- Standardize inputs to the 1-km target grid

PHASE 2:

AOD GAP RECONSTRUCTION

- Identify cloud-induced MAIAC AOD gaps
- Adapt the existing imputation framework to daily resolution
- Develop and evaluate the model for India
- Generate continuous daily 1-km AOD

PHASE 3:

$PM_{2.5}$ ESTIMATION

- Apply the established AOD → $PM_{2.5}$ physical conversion
- Integrate atmospheric aerosol information
- Generate daily 1-km surface $PM_{2.5}$
- Extend the existing 2000–2019 framework toward 2000–2025

PHASE 4:

VALIDATION & EVALUATION

- Validate AOD using AERONET
- Evaluate $PM_{2.5}$ using CPCB observations
- Assess performance across seasons and regions
- Quantify uncertainty and identify limitations

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[7] Central Pollution Control Board (CPCB), India, “National Air Quality Monitoring Network.”